



Prompt-driven UAV Mission Planning

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Motivation



UAV mission planning is complex and largely manual



Requires technical expertise from operators



Largely time-consuming process



Mission planning prone to human error

There is a clear need for an intelligent, language-driven UAV mission planning system.

Problem Statement

"To develop a system that converts natural language mission descriptions into executable UAV flight plans for autonomous navigation."

Natural Language
Input

Intelligent
Processing

Executable
Flight Plans

Project Objectives

01

Convert natural language prompts into structured mission parameters

02

Generate waypoint-based UAV mission plans automatically

03

Integrate with ArduPilot autopilot using MAVLink protocol

04

Validate system in simulation environment

Literature Survey — Evolution of UAV Mission Planning



↑ *Flow: Increasing autonomy and natural language capability over time*

Each era addressed specific limitations of the previous

Gaps remain in real-world integration and MAVLink support

Traditional UAV Mission Planning

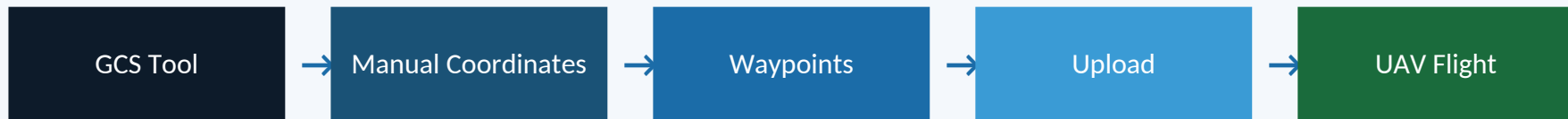
Approach

- Waypoint-based planning using GCS tools (e.g., Mission Planner)
- Manual input of GPS coordinates and altitudes
- Operator-driven flight path configuration

Limitations

- Requires significant domain expertise
- Highly error-prone with manual data entry
- Not scalable for dynamic or complex missions

Typical Workflow:



Algorithmic Path Planning

Key Algorithms:

A*

Optimal heuristic search algorithm

Dijkstra

Shortest path graph traversal

RRT

Rapidly-exploring Random Trees

Potential Fields

Attraction/repulsion force model

Focus Areas

- Optimal path computation
- Obstacle avoidance in structured environments
- Primarily offline/pre-planned navigation

Key Limitation

- Requires structured, numeric input
- No natural language understanding
- Cannot interpret operator intent directly

Learning-based Approaches

Reinforcement Learning

Agent learns optimal policy through environment interaction and reward signals

Neural Networks

Deep learning models for pattern recognition and decision-making in navigation

Multi-Agent UAV Systems

Coordinated learning across multiple UAV agents for complex missions

✓ Advantage

- Adaptive behavior
- Handles dynamic environments

✗ Limitation

- No natural language understanding
- Cannot interpret mission intent



Need: Language understanding layer

Natural Language Interfaces for UAVs

Rule-based Systems

- Predefined command-response mappings
- Keywords trigger specific actions
- Simple and fast for known commands

Voice Command Systems

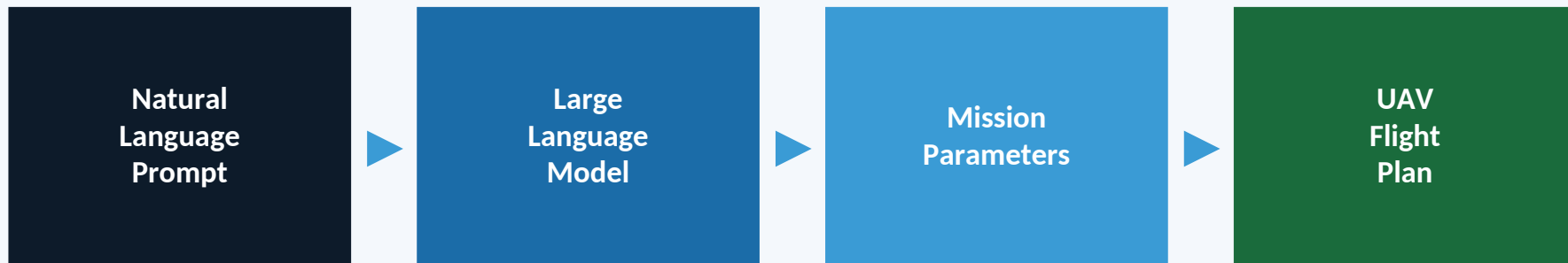
- Speech-to-text with intent parsing
- Used in consumer drones (e.g. DJI Mavic)
- Real-time interaction capability

Limitations of Existing NL Interfaces

- Limited vocabulary — only fixed commands supported
- Cannot handle complex, multi-step or spatial instructions
- No semantic understanding of mission intent
- Fails on ambiguous or varied phrasings

LLM-based UAV Systems

Processing Pipeline:



Advantages

- Handles free-form, varied natural language
- Flexible — no fixed command vocabulary
- Context-aware interpretation of mission goals
- Rapidly improving with foundation model advances

Notable Systems

- ChatGPT-based drone controllers (GPT-4)
- Code-as-plans (inner monologue) systems
- Multi-modal robot command interfaces
- Foundation model-driven path executors

Limitations of Existing Work

Cloud Dependency



Most LLM systems require internet and paid APIs — not deployable offline

Limited Real-world Validation

Mostly simulation-only; minimal testing on actual UAV hardware

No MAVLink Integration

Lack of direct interface with ArduPilot / MAVLink autopilot stack

High Latency



Cloud round-trips introduce delays unsuitable for time-critical UAV tasks

Control vs Planning Focus



Prior work focuses on low-level control rather than high-level mission planning

Project Scope

✓ In Scope

- Focus on mission-level planning
- Grid and waypoint mission types
- ArduPilot / MAVLink integration
- Simulation-based validation

Out of Scope

- Obstacle avoidance
- Real-time mission replanning
- Multi-UAV coordination
- Computer vision integration
- Hardware-level control loops

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Thank You
